

AKSHAYA JAYAKANTH

📍 Providence, RI | ✉️ akshaya_jayakanth@brown.edu | [in](#) LinkedIn | [G](#)itHub

EDUCATION

Brown University – Providence, Rhode Island

Aug 2025 – May 2027

ScM in Data Science

Relevant Coursework: Hands-on Data Science, Data Engineering, Fairness in Automated Decision Making, Statistical Learning.

SRM Institute of Science and Technology – Chennai, India

Sept 2021 – June 2025

B.Tech in Computer Science Engineering, specialization in Big Data Analytics

Relevant Coursework: Data Structures & Algorithms, OOP, Probability & Queuing Theory, Calculus, Linear Algebra, DBMS.

PROFESSIONAL EXPERIENCE

Brown University – Center for Career Exploration, Student Research Assistant, Providence, RI

Jun 2026 – Present

- Architected a **unified relational schema** for the CR@B project, normalizing career engagement and outcome data from multiple institutional sources to enforce referential integrity; authored **analytical SQL queries** including window functions and multi-table joins to power downstream visualization pipelines.
- Engineered **modular ETL pipelines** in Python and SQL on Brown's **OSCAR HPC cluster**, automating multi-source ingestion and transforming raw records into a standardized schema, eliminating manual data reconciliation for the research team.

In22labs (Unwind Learning Labs Pvt Ltd), AI/ML Intern, Chennai (Hybrid)

Aug 2024 – Jan 2025

- Engineered a **production document QA system** with OCR-based multi-format ingestion (PDF, TXT, PNG, JPG), serving inference via a **FastAPI** endpoint, and migrating the backend from OpenAI APIs to locally-deployed **Llama 3.2 via Ollama** after benchmarking Gemma and Mistral on retrieval accuracy and latency, cutting inference costs to zero.
- Identified, proposed, and closed a formal **MoU between In22labs and SRM University**, then designed and ran end-to-end recruitment and onboarding for both intern cohorts, **growing the team by 17 interns** while completing final-year undergraduate coursework.

Agasthya Data Solution, Data Analytics Intern, Hyderabad

Jun 2023 – Jul 2023

- Engineered **time-series feature extraction** pipelines across 5 years of daily Nifty50 price and volume data for 50 companies in Python and SQL, identifying momentum and sector-level patterns that directly seeded a published paper on **deep reinforcement learning for portfolio optimization** (IJSREM).

PROFESSIONAL PRACTICUM

Teacher Education Research (Annenberg Institute of Education) - Capstone, Brown University

Jun 2026 – Present

- Applying **topic modeling and transformer-based text classification** pipelines to tens of thousands of qualitative teacher induction narratives linked to **Connecticut state administrative records**, quantifying thematic predictors of early-career teacher attrition across school and district contexts.

PROJECTS

- **Early Prediction of Movie Box Office Success:** Engineered time-aware Wikipedia activity features and compared distributional model specifications, selecting a **Gamma GLM** with log-log transformation to handle skewed revenue; achieved cross-validated R^2 of **0.78** via 10-fold CV across 312 films, identifying page views and theater count as **statistically significant** dominant predictors.
- **In-Hospital Mortality Prediction (SUPPORT2 Dataset):** Trained and compared five ML models on clinical records from **9,105 critically ill patients** via 5-fold stratified cross-validation in Scikit-learn; achieved **0.916 AUC** with Gradient Boosting, pinpointed acute physiology severity (apvssst) as dominant predictor via **SHAP**; shipped with 40 unit tests and CI/CD pipeline.
- **Anomaly Detection in S&P 500 (Deep Learning):** Trained a **TensorFlow autoencoder** on historical S&P 500 data to flag irregular trading patterns using reconstruction error thresholding; achieved **80% detection accuracy**; validated with 21 unit tests.
- **Equity Portfolio Analysis Tool:** Engineered a real-time pipeline ingesting 1,500 trading days across **50+ stocks** via the yfinance API; implemented momentum-based selection and benchmarked risk-adjusted performance (CAGR, volatility, Sharpe ratio) in a **Streamlit** dashboard; 34 unit tests with CI/CD pipeline.
- **Vision-Based Query Retrieval System:** Engineered a document QA system with **OCR-based multi-format ingestion** (PDF, TXT, PNG, JPG); migrated from OpenAI APIs to locally-deployed Llama 3.2 via Ollama after benchmarking Gemma and Mistral on retrieval accuracy and latency, cutting inference costs to zero.

PUBLICATIONS & PATENT

- **AI-Driven Liver Disease Diagnosis: Multi-Modal Fusion of Medical Imaging and Clinical Reports (Patent No. 202541059774)** – Designed a multi-modal AI system fusing EHR records, medical imaging, and NLP pipelines for automated liver disease detection; achieved **85–93% precision**.
- **Enhancing Portfolio Optimisation with Deep Reinforcement Learning (IJSREM)** – Designed a DRL framework using DQN, PPO, and TRPO on NIFTY50 data with Sharpe ratio as the reward signal; outperformed Mean-Variance Optimisation in volatile, nonlinear market conditions.
- **Real-Time Personality Assessment via Big Five Traits (Research Square)** – Built an NLP pipeline using embeddings, sentiment scoring, and behavioral features trained on a **4GB dataset** for automated personality evaluation in HR workflows.

SKILLS

Languages

Python, R, SQL, C, C++

ML & AI

Machine Learning, Deep Learning, NLP, Statistical Modeling, Predictive Modeling

Libraries & Frameworks

Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, FastAPI, Streamlit

Databases & Data Engineering

MySQL, PostgreSQL, MongoDB, ETL Pipelines, Data Warehousing

Tools & Platforms

Git, Jupyter, Tableau, Power BI, Matplotlib, Seaborn, Linux, OSCAR (Brown HPC)